Wentao Song

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I am a wildlife ecologist and applied statistician with comprehensive training in biology and ecology for wildlife management. My doctoral research focuses on modeling the spatiotemporal dynamics of North American breeding bird populations. I have contributed to a range of projects involving birds, rodents, deer, and primates. My interest is in applying statistical models and machine learning methods to analyze and predict population dynamics, with the goal of supporting wildlife management and conservation.

EDUCATION

Mississippi State University Mississippi, US
PhD. Natural Resources, Wildlife concentration 2019 - 2024
Advisor: Dr. Guiming Wang

Capital Normal University Beijing, China
MSc. Zoology 2015 - 2018
Advisor: Dr. Zihui Zhang

Institute of Zoology, Chinese Academy of Sciences Beijing, China
Joint Master Program. Population ecology of small mammals 2015 - 2018
Advisor: Dr. Xinrong Wan

China Agricultural University Beijing, China
BSc(Agr.) Aquaculture 2011 - 2015

EMPLOYMENT EXPERIENCE

Mississippi State University Mississippi, US
Teaching assistant 2019.08 - 2024.12
Assisted instructors in preparing lectures and labs for undergraduate students.

Independent Statistical Consultant Florida, US
Statistical consulting and data analysis service 2025.01 - Now

SKILLS

QUANTITATIVE SKILLS

- Time series analysis
- Spatial modeling
- Bayesian hierarchical modeling
- Structured decision making
- Coding with R, Python, Julia, and SAS
- Bayesian analysis with PyMC3, WinBUGS, JAGS, and Stan

TEACHING

- High school teaching certification (biology and physics) in China
- Ecology, Population Ecology, and Community Ecology
- Ornithology and Mammalogy
- Ecological Modeling

FIELDWORK & LABWORK

Small mammal trapping
Primate movement tracking
Deer spotlight survey and distance sampling
Bird banding
Polymerase Chain Reaction and Electrophoresis

LANGUAGES

Advanced speaker of Mandarin and English
Pre-intermediate speaker of Japanese and French

COURSES ASSISTED

Applied Aquatic and Terrestrial Ecology (WFA3133, Dr. Sandra Correa). Fall 2023.
Mammalogy (WFA 4433, Dr. Christopher Ayers). Fall 2022.
Ornithology (WFA 4443, Dr. Christopher Ayers). Spring 2021.
Wildlife and Fishery Practice (WFA4473, Dr. John Davis). Spring 2020.

INVITED LECTURES

Responses of bird populations to climatic changes. Applied Ecology. MSU, Fall 2023.
Series Lectures of Conservation Biology:
(1)The IUCN red list of threatened species: How it's being used.
(2)Habitat destruction: Death by a thousand cuts.
(3)Reintroduction and rewilding as conservation strategies.
(4)Climate change: Implications for biodiversity.
Current Topics in Conservation Biology. MSU, Fall 2021.
Clustering Methods. Quantitative and Statistical Ecology. MSU, Spring 2021.
Plotting with Matplotlib. Python Club. MSU, Fall 2020.

CONFERENCE PRESENTATIONS

Variations in the responses of North American breeding bird populations to climatic changes. The Wildlife Society's 30th Annual Conference, Louisville, KY, 2023.

Variations in the responses of North American breeding bird populations to climatic changes. The Wildlife Society Mississippi Chapter Annual Meeting, Jackson, MS, 2023.

Population dynamic patterns of North American breeding birds under global changes. The Wildlife Society's 29th Annual Conference, Spokane, WA, 2022.

Machine Learning for Identifying Population Dynamic Patterns of North American Breeding Birds. The Wildlife Society Mississippi Chapter Annual Meeting (online), 2022.

The wave: a dispersal hypothesis about space dynamics of Brandt's vole. The 8th International Symposium of Integrative Zoology. Xilinhaote, China, 2016.

Publications

Song, W., Kouba, A.J., Burger, L.M., Burger, L.W., Colvin, M.E., Wang, G. Interactions of life history traits, density dependence and responses to climate changes on long-term dynamic patterns of North American breeding bird populations. (In prep for Global Change Biology)

Shan, X., **Song, W.**, Strickland, B., Demarais, S., Beightol, C.V., Wang, G. (2026). Fine-scale spatial genetic structure of white-tailed deer in a national park in the southeastern United States. Mammalian Biology, .

Yang, H., Li, S., Hou, R., **Song, W.**, Fu, Y., Li, Y., ... & Li, B. (2023). Three-dimensional assessment of movement patterns of Sichuan snub-nosed monkeys affected by habitat structure in temperate forests. Zoological Research, 44(2), 361.

Song, W., Wang, Y., Zhang, X., Zhang, W., … & Wan, X. (2017). Influence of Group Size and Foraging Distance on Vigilance Frequency of Brandt's Vole (*Lasiopodomys brandtii*) in Food Storing Period. Chinese Journal of Zoology, 52(5): 754-760.

Song, W., Wang, Y., Sai, N., … & Zhang, Z. (2016). Numerical Response of Hawks Density to the Rodents Density in Typical Steppe. Chinese Journal of Zoology, 51(4): 529-535.

REFEREES

Dr. Guiming Wang (PhD advisor)
Professor, Mississippi State University
guiming.wang@msstate.edu

Dr. Xueyan Shan (Project advisor)
Associate Research Professor, Mississippi State University
xshan@bch.msstate.edu

Dr. Sandra Correa (TA advisor)
Associate Professor, Mississippi State University
sbc257@msstate.edu

Dr. Xinrong Wan (Master's advisor)
Associate Professor, Institute of Zoology, Chinese Academy of Sciences
wanxr@ioz.ac.cn